

MEDICAL MANAGEMENT SYSTEM

Course name: Database Management System

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1.Objectives

- ❑ **Efficient Patient Management:**
 - To provide a systematic and centralized way of registering and maintaining patient information, including personal details, medical history, and treatment records.
 - To reduce errors and redundancy caused by manual record-keeping.
- ❑ **Doctor and Staff Scheduling:**
 - To manage doctor profiles, specializations, and daily/weekly schedules efficiently.
 - To allow staff to easily access doctor availability for appointment scheduling.
- ❑ **Appointment Management:**
 - To automate appointment booking, rescheduling, and cancellation.
 - To notify patients and doctors of upcoming appointments through reminders, reducing waiting times and missed consultations.
- ❑ **Billing and Payment Tracking:**
 - To generate accurate and professional invoices for services rendered.
 - To track payments (cash, card, insurance) efficiently and reduce billing errors.
 - To maintain a detailed financial record for hospital administration and audits.
- ❑ **Pharmacy and Medicine Inventory Management:**
 - To maintain a real-time inventory of medicines and supplies.
 - To manage prescriptions and ensure correct medicine dispensing.
 - To automatically alert when stocks are low or medicines are near expiry.
- ❑ **Reports and Analytics:**
 - To provide comprehensive reports on patient visits, doctor performance, revenue, and medicine usage.
 - To assist hospital management in decision-making and operational improvements.
- ❑ **Security and Role-Based Access:**
 - To ensure data confidentiality through secure login credentials.
 - To assign access rights based on roles (Admin, Doctor, Receptionist, Pharmacist, Accountant) ensuring that sensitive information is protected.

2.Problem Statement:

- ❑ **Inefficient Patient Management:** Patient records are scattered and difficult to access, causing delays in retrieving medical history and treatment information.
- ❑ **Appointment Conflicts:** Manual scheduling often results in double bookings, missed appointments, and long waiting times for patients.
- ❑ **Billing Errors:** Manual billing increases the risk of calculation errors, missing payments, and lack of transparency.
- ❑ **Medicine Stock Issues:** Pharmacy inventory management is often inaccurate, leading to stock shortages, expired medicines, or wastage.
- ❑ **Limited Reporting and Analytics:** Hospital administrators lack real-time data to make informed decisions regarding patient care, staff performance, or financial management.

3.Solve Statement:

- ❑ **Centralized Patient Management:** Store and manage all patient records digitally, ensuring easy access to medical history, treatments, and personal information.
- ❑ **Automated Appointment Scheduling:** Allow patients and staff to book, reschedule, or cancel appointments without conflicts, reducing waiting time and improving patient satisfaction.
- ❑ **Accurate Billing and Payment Tracking:** Generate professional invoices, track all payments, and maintain a transparent financial record for hospital management.
- ❑ **Efficient Pharmacy and Medicine Inventory Management:** Keep real-time records of medicines, manage prescriptions, and provide alerts for low stock or nearing expiry dates.

□ **Comprehensive Reports and Analytics:** Provide administrators with dashboards and reports on patient visits, doctor performance, revenue, and medicine usage to enable data-driven decision-making.

4. Project Scope:

The **Medical Management System (MMS)** is designed to automate hospital operations and improve efficiency, accuracy, and security.

4.1 Functional Scope

- **Patient Management:** Register, update, and maintain patient records with medical history.
- **Doctor Management:** Manage doctor profiles, schedules, and appointments.
- **Appointment Management:** Book, reschedule, or cancel appointments; check doctor availability.
- **Billing & Payment:** Generate invoices, track payments, and maintain financial records.
- **Pharmacy Management:** Manage medicine inventory, prescriptions, and stock alerts.
- **Reports & Analytics:** Generate reports on patients, revenue, staff performance, and medicine usage.

4.2 Non-Functional Scope

- Fast, reliable, and user-friendly system
- Data security, backup, and recovery
- Scalable to handle increasing hospital operations

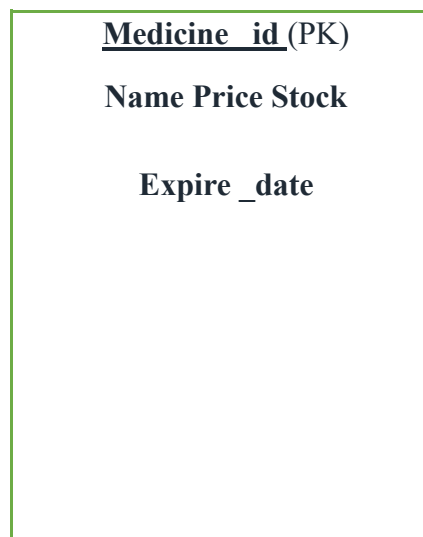
5. Entities

5.1 StrongEntities

Appointment



Medicine



WeakEntities:

Dependent on: **Bill**

Bill

<u>Bill_id</u> (PK)
Patient_id (FK)
Total_amount bil

Bill_Details

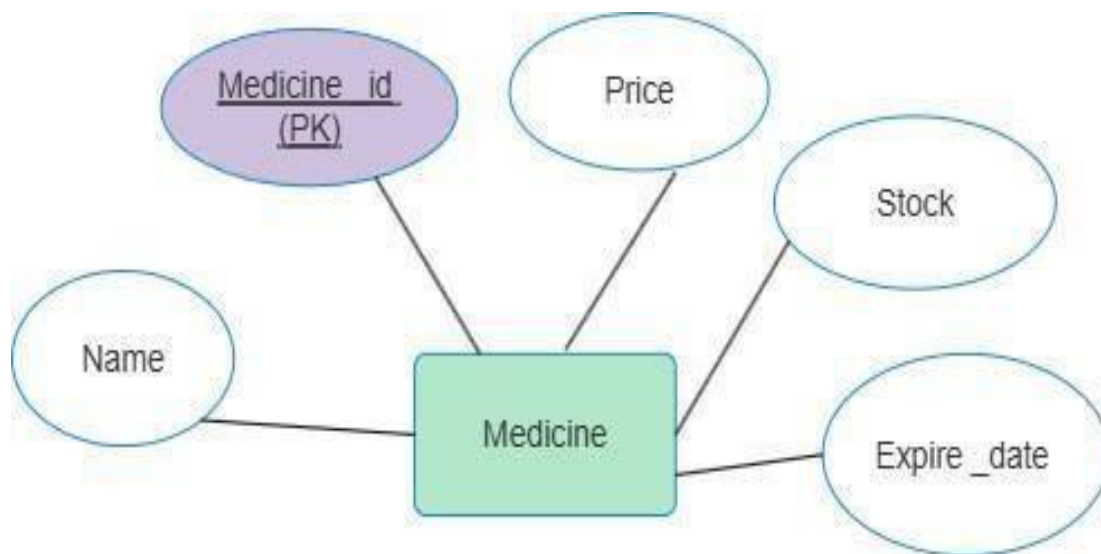
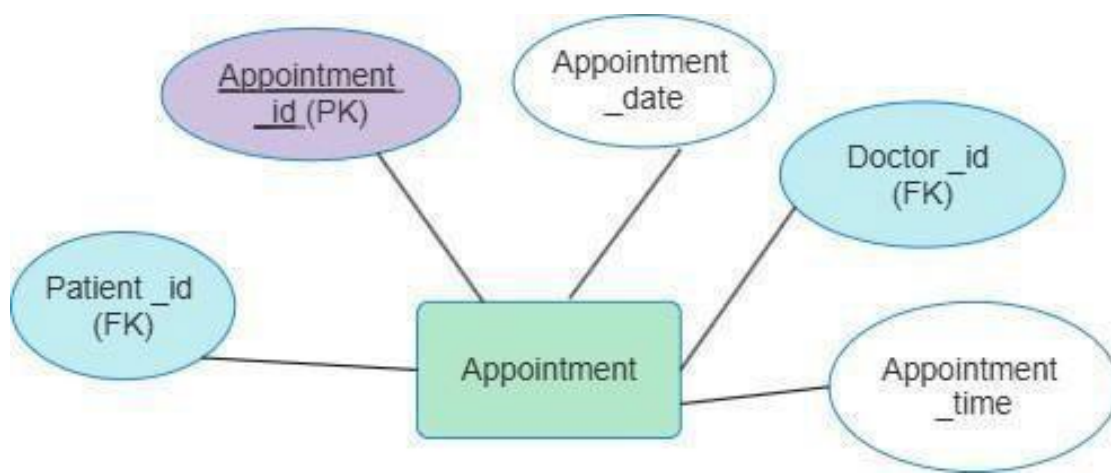
Bill_detail_idname
Bill_id (FK)
referencing Bill)
Item_type
Item_Amount

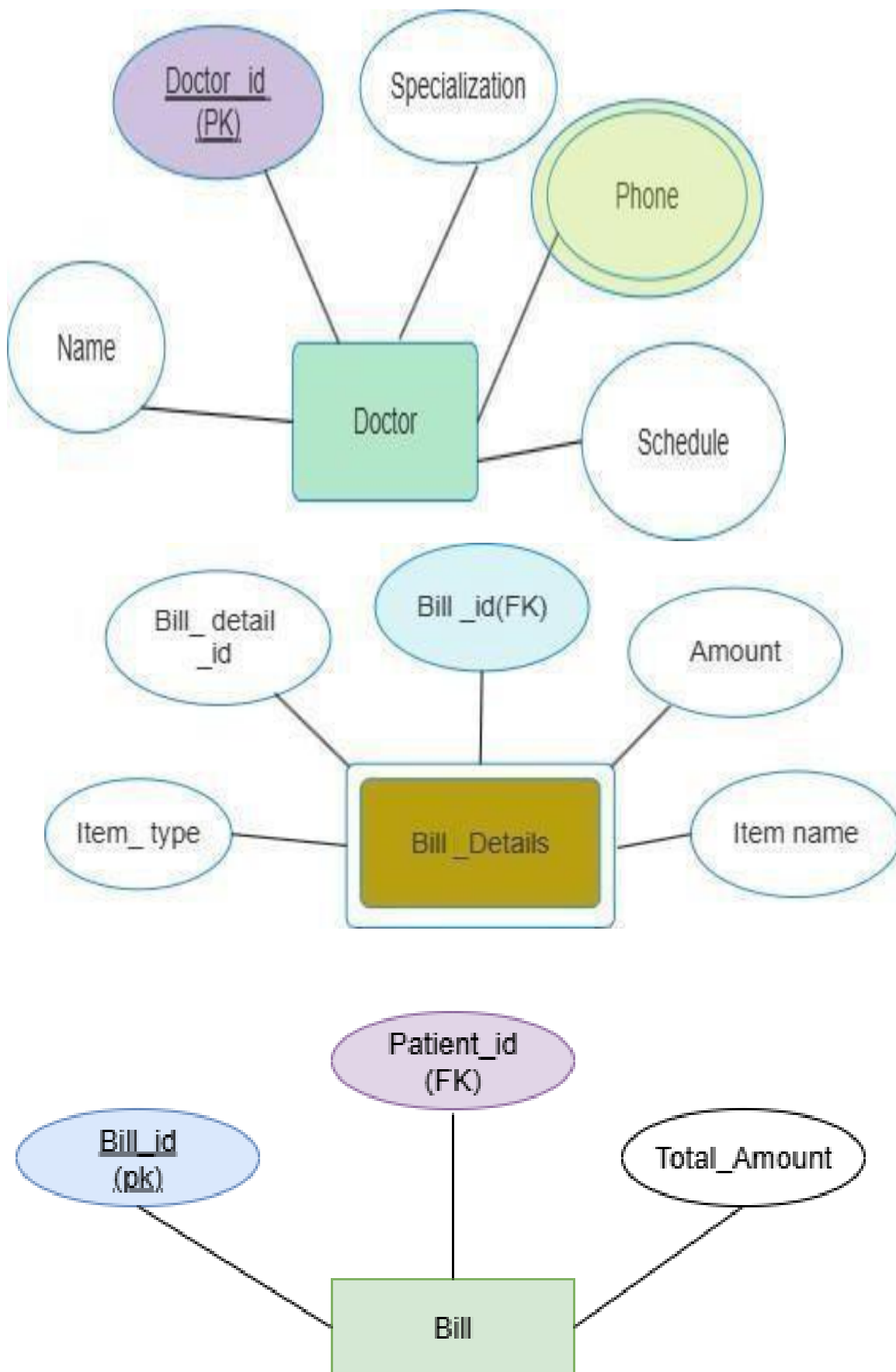
Doctor

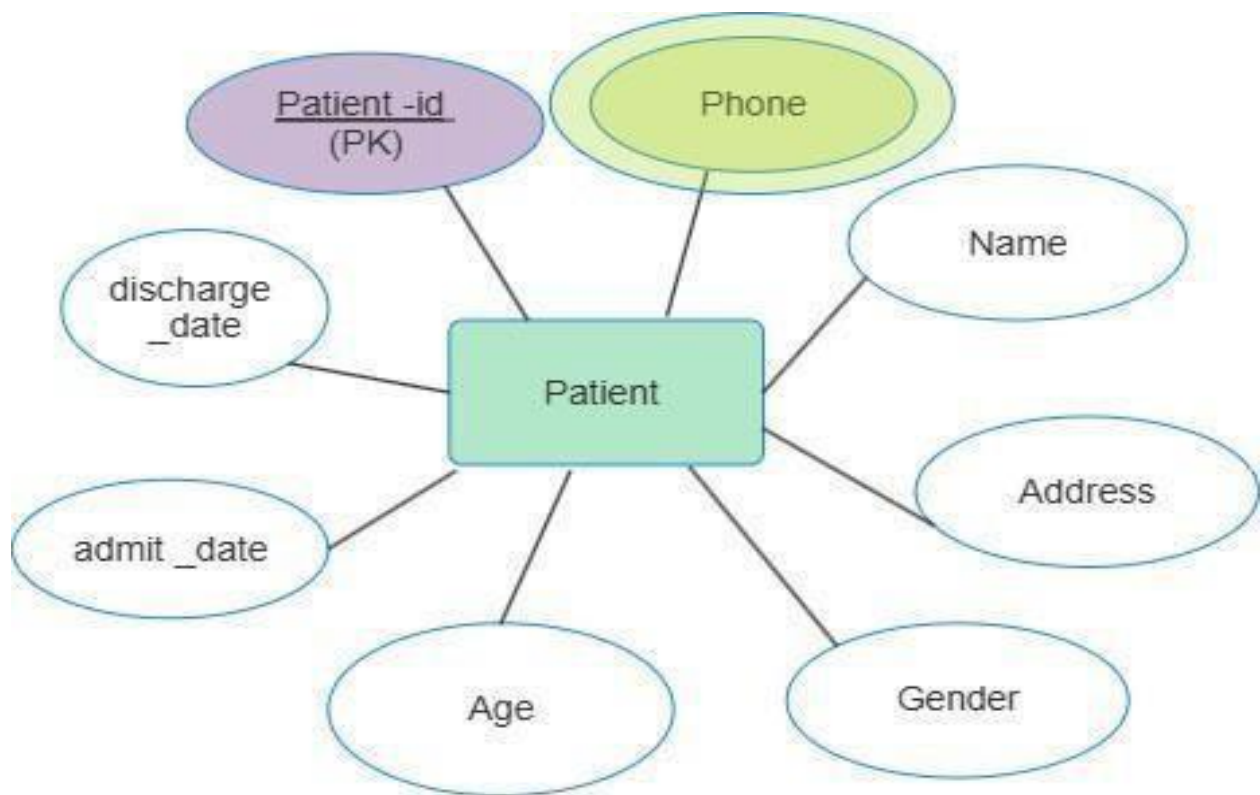
<u>Doctor_id</u> (PK)
Name
Specialization
Phone Schedule

Patient

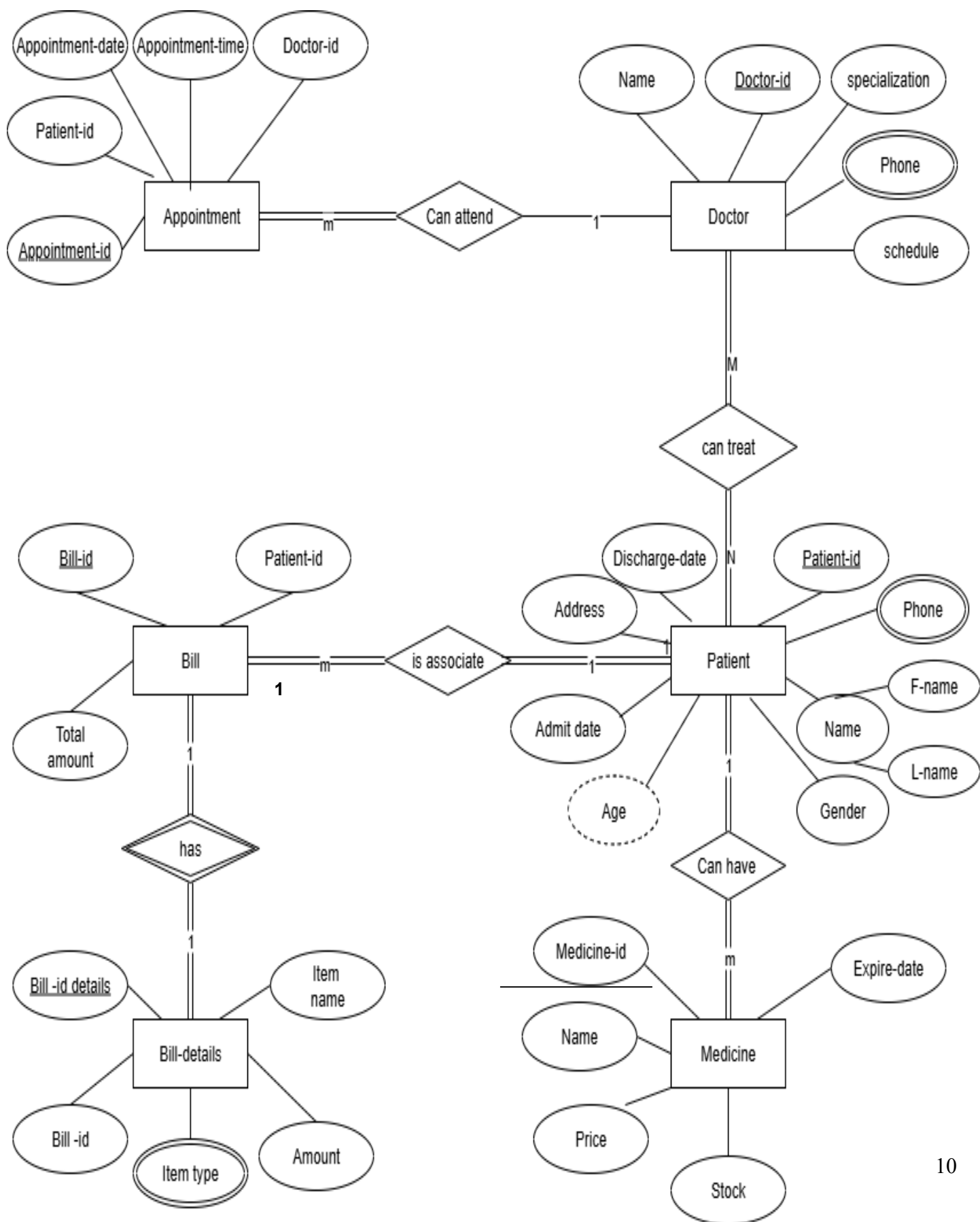
<u>Patient_id</u> (PK)
Name
age
gender
phone
address
admit_date
discharge_date







6 .ER Diagram:



7.ER to relation mapping

Appointment:

Appointment-id (pk)	Patient-id (FK)	Appointr -	Doctor-id (FK)	Appointmenttime
------------------------	-----------------	------------	-------------------	-----------------

Doctor:

Doctor-id (PK)	Specialization	Phone	Name	Schedule
----------------	----------------	-------	------	----------

Patient:

Patient-id	Phone	Name	Adress	Gender	Age	Admitdate	Dischargedate
------------	-------	------	--------	--------	-----	-----------	---------------

Medicine:

Medicine-id (PK)	Price	Name	Stock	Expire-date
------------------	-------	------	-------	-------------

Bill:

Bill-id (PK)	Patient-id (FK)	Total-amount
--------------	-----------------	--------------

Bill-Details:

Bill-details-id (PK)	Bill-id (FK)	Amount	Item name	Item-type
-------------------------	--------------	--------	-----------	-----------

7.1 Relationships:

One Doctor can attend many Appointments (1-to-many). One Patient can have many Prescriptions (1-to-many).

One Bill is associated with one Patient. (1 to 1)

Patients can be treated by multiple doctors.

8. Implementation

Implementation is the phase where the actual development and working system is realized. It includes modules, database integration, frontend-backend interaction, and testing.

9. Testing:

9.1 Usability Testing

- Interface should be **user-friendly**.
- Forms should have **validation** (e.g., phone number must be 11 digits, email valid).
- Dashboard should **display important info** (today's appointments, total patients, revenue).

9.2 Security Testing

- Login requires **username & password**
- Role-based access:
 - Admin → Full access
 - Doctor → Only patient info and appointments
 - Receptionist → Add patients, book appointments
- Check SQL Injection: All input fields sanitized

10. Sql Query:

```
1 • create database medical;
2 • CREATE TABLE Doctor (
3     Doctor_id INT PRIMARY KEY,
4     Specialization varchar(100),
5     Phone varchar(15),
6     Name varchar(100),
7     Schedule varchar(100)
8 );
9 • INSERT INTO Doctor (Doctor_id, Specialization, Phone, Name, Schedule) VALUES
10 (1, 'Cardiologist', '01711----', 'Dr. Rahman', 'Sun-Tue 10AM-2PM'),
11 (2, 'Neurologist', '01722-----', 'Dr. Karim', 'Wed-Fri 9AM-1PM'),
12 (3, 'Dentist', '0173-----', 'Dr. Afsana', 'Mon-Wed 11AM-3PM'),
13 (4, 'Orthopedic', '0174-----', 'Dr. Hasan', 'Thu-Sat 10AM-4PM');
14 • select * from Doctor;
15 • CREATE TABLE Patient (
16     Patient_id INT PRIMARY KEY,
17     Phone VARCHAR(15),
18     Name VARCHAR(100),
19     Address VARCHAR(150),
20     Gender VARCHAR(10),
21     Age INT,
22     Admit_date DATE,
23     Discharge_date DATE
24 );
25 • INSERT INTO Patient (Patient_id, Phone, Name, Address, Gender, Age, Admit_date, Discharge_date) VALUES
26 (1, '0181-----', 'Rafi Ahmed', 'Dhaka', 'Male', 25, '2025-10-01', '2025-10-05'),
27 (2, '0181-----', 'Sadia Islam', 'Chittagong', 'Female', 30, '2025-09-20', '2025-09-28'),
28 (3, '0181-----', 'Imran Hossain', 'Sylhet', 'Male', 45, '2025-10-10', '2025-10-15'),
29 (4, '01814-----', 'Mitu Akter', 'Khulna', 'Female', 35, '2025-10-03', '2025-10-09');
30 • select * from Patient;
```

```

31 • CREATE TABLE Appointment (
32     Appointment_id INT PRIMARY KEY,
33     Patient_id INT,
34     Appointment_date DATE,
35     Doctor_id INT,
36     Appointment_time TIME,
37     FOREIGN KEY (Patient_id) REFERENCES Patient(Patient_id),
38     FOREIGN KEY (Doctor_id) REFERENCES Doctor(Doctor_id)
39 );
40 • INSERT INTO Appointment (Appointment_id, Patient_id, Appointment_date, Doctor_id, Appointment_time) VALUES
41     (1, 1, '2025-10-02', 1, '10:30:00'),
42     (2, 2, '2025-10-04', 3, '11:00:00'),
43     (3, 3, '2025-10-12', 2, '09:45:00'),
44     (4, 4, '2025-10-06', 4, '13:15:00');
45 • select * from Appointment;
46 • CREATE TABLE Medicine (
47     Medicine_id INT PRIMARY KEY,
48     Price DECIMAL(10,2),
49     Name VARCHAR(100),
50     Stock INT,
51     Expire_date DATE
52 );
53 • INSERT INTO Medicine (Medicine_id, Price, Name, Stock, Expire_date) VALUES
54     (1, 120.50, 'Paracetamol', 100, '2026-05-01'),
55     (2, 250.00, 'Amoxicillin', 80, '2026-02-15'),
56     (3, 80.75, 'Cetirizine', 150, '2026-07-10'),
57     (4, 300.00, 'Omeprazole', 60, '2025-12-20');

```

```

58 • select * from Medicine;
59 • CREATE TABLE Bill (
60     Bill_id INT PRIMARY KEY,
61     Patient_id INT,
62     Total_amount DECIMAL(10,2),
63     FOREIGN KEY (Patient_id) REFERENCES Patient(Patient_id)
64 );
65 • INSERT INTO Bill (Bill_id, Patient_id, Total_amount) VALUES
66     (1, 1, 1500.00),
67     (2, 2, 2300.50),
68     (3, 3, 1750.00),
69     (4, 4, 1980.75);
70 • select * from Bill;
71 • CREATE TABLE Bill_Details (
72     Bill_details_id INT PRIMARY KEY,
73     Bill_id INT,
74     Amount DECIMAL(10,2),
75     Item_name varchar(100),
76     Item_type varchar(50),
77     FOREIGN KEY (Bill_id) REFERENCES Bill(Bill_id)
78 );
79 • INSERT INTO Bill_Details (Bill_details_id, Bill_id, Amount, Item_name, Item_type) VALUES
80     (1, 1, 500.00, 'Consultation', 'Service'),
81     (2, 1, 1000.00, 'Medicine', 'Product'),
82     (3, 2, 1300.50, 'Lab Test', 'Service'),
83     (4, 2, 1000.00, 'Medicine', 'Product'),
84     (5, 3, 750.00, 'Consultation', 'Service'),
85     (6, 3, 1000.00, 'Medicine', 'Product');
86 • select * from Bill_Details;

```

10.1 Doctor Table:

```

CREATE TABLE Doctor (
    Doctor_id INT PRIMARY KEY,
    Specialization varchar (100),
    Phone varchar (15),
    Name varchar (100),
    Schedule varchar (100)
);

INSERT INTO Doctor (Doctor_id, Specialization, Phone, Name, Schedule) VALUES
(1, 'Cardiologist', '01711----', 'Dr. Rahman', 'Sun-Tue 10AM-2PM'),
(2, 'Neurologist', '01722-----', 'Dr. Karim', 'Wed-Fri 9AM-1PM'),
(3, 'Dentist', '0173-----', 'Dr. Afsana', 'Mon-Wed 11AM-3PM'),
(4, 'Orthopedic', '0174-----', 'Dr. Hasan', 'Thu-Sat 10AM-4PM');

```


Output Table:

	Doctor_id	Specialization	Phone	Name	Schedule
▶	1	Cardiologist	01711----	Dr. Rahman	Sun-Tue 10AM-2PM
	2	Neurologist	01722----	Dr. Karim	Wed-Fri 9AM-1PM
	3	Dentist	0173----	Dr. Afsana	Mon-Wed 11AM-3PM
	4	Orthopedic	0174----	Dr. Hasan	Thu-Sat 10AM-4PM
✱	NULL	NULL	NULL	NULL	NULL

10.2 Patient Table:

```
CREATE TABLE Patient (
    Patient_id INT PRIMARY KEY,
    Phone VARCHAR(15),
    Name VARCHAR(100),
    Address VARCHAR(150),
    Gender VARCHAR(10),
    Age INT,
    Admit_date DATE,
    Discharge_date DATE
);
INSERT INTO Patient (Patient_id, Phone, Name, Address, Gender, Age, Admit_date, Discharge_date)
VALUES
(1, '0181-----', 'Rafi Ahmed', 'Dhaka', 'Male', 25, '2025-10-01', '2025-10-05'),
(2, '0181-----', 'Sadia Islam', 'Chittagong', 'Female', 30, '2025-09-20', '2025-09-28'),
(3, '0181-----', 'Imran Hossain', 'Sylhet', 'Male', 45, '2025-10-10', '2025-10-15'),
(4, '01814-----', 'Mitu Akter', 'Khulna', 'Female', 35, '2025-10-03', '2025-10-09');
```

Output Table:

	Patient_id	Phone	Name	Address	Gender	Age	Admit_date	Discharge_date
▶	1	0181-----	Rafi Ahmed	Dhaka	Male	25	2025-10-01	2025-10-05
	2	0181-----	Sadia Islam	Chittagong	Female	30	2025-09-20	2025-09-28
	3	0181-----	Imran Hossain	Sylhet	Male	45	2025-10-10	2025-10-15
	4	01814-----	Mitu Akter	Khulna	Female	35	2025-10-03	2025-10-09
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

10.3 Appointment Table:

```
CREATE TABLE Appointment (
    Appointment_id INT PRIMARY KEY,
    Patient_id INT,
    Appointment_date DATE,
    Doctor_id INT,
    Appointment_time TIME,
```

```

FOREIGN KEY (Patient_id) REFERENCES Patient(Patient_id),
FOREIGN KEY (Doctor_id) REFERENCES Doctor(Doctor_id)
);
INSERT INTO Appointment (Appointment_id, Patient_id, Appointment_date, Doctor_id, Appointment_time)
VALUES
(1, 1, '2025-10-02', 1, '10:30:00'),
(2, 2, '2025-10-04', 3, '11:00:00'),
(3, 3, '2025-10-12', 2, '09:45:00'),
(4, 4, '2025-10-06', 4, '13:15:00');

```

Output Table:

	Appointment_id	Patient_id	Appointment_date	Doctor_id	Appointment_time
▶	1	1	2025-10-02	1	10:30:00
	2	2	2025-10-04	3	11:00:00
	3	3	2025-10-12	2	09:45:00
	4	4	2025-10-06	4	13:15:00
✱	NULL	NULL	NULL	NULL	NULL

10.4 Medicine Table:

```

CREATE TABLE Medicine (
    Medicine_id INT PRIMARY KEY,
    Price DECIMAL(10,2),
    Name VARCHAR(100),
    Stock INT,
    Expire_date DATE
);
INSERT INTO Medicine (Medicine_id, Price, Name, Stock, Expire_date) VALUES
(1, 120.50, 'Paracetamol', 100, '2026-05-01'),
(2, 250.00, 'Amoxicillin', 80, '2026-02-15'),
(3, 80.75, 'Cetirizine', 150, '2026-07-10'),
(4, 300.00, 'Omeprazole', 60, '2025-12-20');

```

Output Table:

	Medicine_id	Price	Name	Stock	Expire_date
▶	1	120.50	Paracetamol	100	2026-05-01
	2	250.00	Amoxicillin	80	2026-02-15
	3	80.75	Cetirizine	150	2026-07-10
	4	300.00	Omeprazole	60	2025-12-20

10.5 Bill Table:

```
CREATE TABLE Bill (  
    Bill_id INT PRIMARY KEY,  
    Patient_id INT,  
    Total_amount DECIMAL(10,2),  
    FOREIGN KEY (Patient_id) REFERENCES Patient(Patient_id)  
);  
INSERT INTO Bill (Bill_id, Patient_id, Total_amount) VALUES  
(1, 1, 1500.00),  
(2, 2, 2300.50),  
(3, 3, 1750.00),  
(4, 4, 1980.75);
```

Output Table:

	Bill_id	Patient_id	Total_amount
▶	1	1	1500.00
	2	2	2300.50
	3	3	1750.00
	4	4	1980.75
▲	NULL	NULL	NULL

10.6 Bill_Details Table:

```
CREATE TABLE Bill_Details (  
    Bill_details_id INT PRIMARY KEY,  
    Bill_id INT,  
    Amount DECIMAL(10,2),  
    Item_name varchar(100),  
    Item_type varchar(50),  
    FOREIGN KEY (Bill_id) REFERENCES Bill(Bill_id)  
);  
INSERT INTO Bill_Details (Bill_details_id, Bill_id, Amount, Item_name, Item_type) VALUES  
(1, 1, 500.00, 'Consultation', 'Service'),  
(2, 1, 1000.00, 'Medicine', 'Product'),  
(3, 2, 1300.50, 'Lab Test', 'Service'),  
(4, 2, 1000.00, 'Medicine', 'Product'),  
(5, 3, 750.00, 'Consultation', 'Service'),  
(6, 3, 1000.00, 'Medicine', 'Product');
```

Output Table:

	Bill_details_id	Bill_id	Amount	Item_name	Item_type
▶	1	1	500.00	Consultation	Service
	2	1	1000.00	Medicine	Product
	3	2	1300.50	Lab Test	Service
	4	2	1000.00	Medicine	Product
	5	3	750.00	Consultation	Service
	6	3	1000.00	Medicine	Product

11. Query

Query 1: **Find medicines that will expire before June 2026**

```
SELECT Name, Expire_date
FROM Medicine
WHERE Expire_date < '2026-06-01';
```

Output Table:

	Name	Expire_date
▶	Paracetamol	2026-05-01
	Amoxicillin	2026-02-15
	Omeprazole	2025-12-20

Query 2: **Count how many patients are male and female**

```
SELECT Gender, COUNT(*) AS Total
FROM Patient
GROUP BY Gender;
```

Output Table:

	Gender	Total
▶	Male	2
	Female	2

Query 3: **Show average, maximum, and minimum medicine price**

```
SELECT
    AVG(Price) AS Average_Price,
```

```

MAX(Price) AS Highest_Price,
MIN(Price) AS Lowest_Price
FROM Medicine;

```

Output Table:

	Average_Price	Highest_Price	Lowest_Price
▶	187.812500	300.00	80.75

Query 4: Show patients whose total bill is greater than 1700

```

SELECT
    Patient.Name,
    Bill.Total_amount
FROM Bill
INNER JOIN Patient ON Bill.Patient_id = Patient.Patient_id
WHERE Bill.Total_amount > 1700;

```

Output Table:

	Name	Total_amount
▶	Sadia Islam	2300.50
	Imran Hossain	1750.00
	Mitu Akter	1980.75

Query 5: Find total bill amount for each patient

```

SELECT
    Patient.Name AS Patient_Name,
    Bill.Total_amount
FROM Bill
INNER JOIN Patient ON Bill.Patient_id = Patient.Patient_id;

```

Output Table:

	Patient_Name	Total_amount
▶	Rafi Ahmed	1500.00
	Sadia Islam	2300.50
	Imran Hossain	1750.00
	Mitu Akter	1980.75

Query 6: Display each bill's items (Join Bill and Bill_Details)

```
SELECT
    Bill.Bill_id,
    Patient.Name AS Patient_Name,
    Bill_Details.Item_name,
    Bill_Details.Item_type,
    Bill_Details.Amount
FROM Bill
INNER JOIN Bill_Details ON Bill.Bill_id = Bill_Details.Bill_id
INNER JOIN Patient ON Bill.Patient_id = Patient.Patient_id;
```

Output Table:

	Bill_id	Patient_Name	Item_name	Item_type	Amount
▶	1	Rafi Ahmed	Consultation	Service	500.00
	1	Rafi Ahmed	Medicine	Product	1000.00
	2	Sadia Islam	Lab Test	Service	1300.50
	2	Sadia Islam	Medicine	Product	1000.00
	3	Imran Hossain	Consultation	Service	750.00
	3	Imran Hossain	Medicine	Product	1000.00

Query 7: Show the number of appointments per doctor

```
SELECT
    Doctor.Name AS Doctor_Name,
    COUNT(Appointment.Appointment_id) AS Total_Appointments
FROM Appointment
INNER JOIN Doctor ON Appointment.Doctor_id = Doctor.Doctor_id
GROUP BY Doctor.Name;
```

Output Table:

	Doctor_Name	Total_Appointments
▶	Dr. Rahman	1
	Dr. Karim	1
	Dr. Afsana	1
	Dr. Hasan	1

Query 8: Find the total billing amount of all patients combined

```
SELECT SUM(Total_amount) AS Total_Hospital_Income  
FROM Bill;
```

Output Table:

Total_Hospital_Income
7531.25

Query 9: Display all doctors who work on Wednesday

```
SELECT Name, Specialization, Schedule  
FROM Doctor  
WHERE Schedule LIKE '%Wed%';
```

Output Table:

Name	Specialization	Schedule
Dr. Karim	Neurologist	Wed-Fri 9AM-1PM
Dr. Afsana	Dentist	Mon-Wed 11AM-3PM

Query 10: Display all patients and their assigned doctor

```
SELECT  
    Patient.Name AS Patient_Name,  
    Doctor.Name AS Doctor_Name  
FROM Patient  
LEFT JOIN Appointment ON Patient.Patient_id = Appointment.Patient_id  
LEFT JOIN Doctor ON Appointment.Doctor_id = Doctor.Doctor_id;
```

Output Table:

Patient_Name	Doctor_Name
Rafi Ahmed	Dr. Rahman
Sadia Islam	Dr. Afsana
Imran Hossain	Dr. Karim
Mitu Akter	Dr. Hasan

12. Project Overview

□ It maintains all essential hospital data including **Doctors, Patients, Appointments, Medicines, and Billing information.**

- The system helps to manage daily hospital activities efficiently and accurately.
- Doctors' details such as **specialization, schedule, and contact information** are stored in the system.
- Patient information including **admission, discharge, gender, and address** is properly recorded.
- Appointments between doctors and patients are scheduled and tracked.
- The system keeps records of available medicines, stock quantities, and expiry dates.

13. Conclusion:

This SQL-based **Medical Management System** project was designed to efficiently manage various aspects of a healthcare facility's data using a **relational database model**. The project included multiple interconnected entities such as **Patients, Doctors, Appointments, Prescriptions, Medicines, Bills**, and their respective relationships.